Using radar-optical disagreement to guide flood annotation

Research proposal for CROMA and Sen1Floods11

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The idea is to turn S1-S2 disagreement into a decision about where human annotation is most valuable. Instead of asking only which model scores highest, ask:

With a limited budget for correcting flood labels, which scenes should an analyst inspect to improve performance on future events?

This connects directly to data scarcity. The contribution would depend on demonstrating that the selection strategy works better than established alternatives, not merely proposing disagreement as a score.

1. Why the current results motivate the question

The S2 model performs better overall on the main test, but the joint model improves some Bolivia metrics. This suggests that the information supplied by the sensors has different value across conditions. Aggregate scores do not establish whether their disagreement is useful.

Possible cases:

- S1 misses water that S2 detects: radar ambiguity, small structures, or a weaker learned representation.
- S2 misses water that S1 detects: optical ambiguity or degraded optical observations.
- Both detect different boundaries: resolution effects, acquisition timing, or uncertain annotation.
- Both confidently agree on an incorrect prediction: a shared failure that disagreement cannot detect.

The research starts by distinguishing these cases. A large disagreement score is not automatically evidence of an informative training example.

2. First establish whether disagreement predicts errors

Use the existing models for an exploratory analysis. For each aligned pixel x, obtain predicted water probabilities p_S1(x) and p_S2(x). A simple disagreement score is:

$$D(x) = |p_{S1}(x) - p_{S2}(x)|$$

A score near zero means similar probabilities; a score near one means strong disagreement. Ask:

- Are high-disagreement pixels more frequently wrong against the hand labels?
- Does disagreement reveal errors missed by S2's own uncertainty?
- Does it mostly identify boundaries?
- Does its usefulness change by event or water category?

The key comparison is single-model uncertainty. If S2 uncertainty identifies the same errors equally well, adding an S1 model for annotation selection may provide little practical benefit.

Perform exploratory analysis on training/validation data. Already-inspected test results may motivate the study, but repeated design decisions based on them weaken their role as an independent final evaluation.

3. Make the annotation task concrete

For a first experiment, select whole chips rather than individual pixels. This is easier to implement using the existing hand-labelled data and easier to explain. Convert pixel scores into chip scores carefully:

- Mean disagreement may favour scenes containing extensive ambiguous land.
- Maximum disagreement may select a chip because of one anomalous pixel.
- Averaging a fixed upper fraction of disagreement values is a candidate, with settings chosen on development data.

Start with one simple, predefined aggregation. Avoid searching many scoring rules and reporting only the winner.

Diversity also matters: ten adjacent chips showing the same river may be less useful than ten different environments. Compare disagreement alone with a version that discourages redundant selections.

4. Simulate a real annotation budget

Hide most of the existing hand labels from training and selection. An illustrative pilot:

- Initial labelled set: 20 chips from available training events.
- Candidate pool: remaining hand-labelled training chips, with their labels hidden.
- Selection round: each strategy chooses 10 additional chips.
- Annotation: reveal the selected chips' existing labels.
- Learning: update the model using the expanded labelled set.
- Evaluation: measure performance on fixed validation data and untouched test events.

Repeat rounds to produce learning curves. These numbers are illustrative; final budgets should reflect available chip counts and diversity. Every strategy starts with the same initial labelled set and receives the same budget.

Repeat with several initial subsets because, for a small seed set, the choice of flood events can dominate results. Keep whole test events excluded from both training and selection.

5. Compare against alternatives that could defeat the proposal

Give simple methods a fair chance:
- Random selection: does sophisticated selection help at all?
- S2 uncertainty: does radar supply additional selection information?
- Representation diversity: is covering different environments sufficient?
- Predicted-water coverage: does the strategy simply select more water?
- S1-S2 disagreement: the central candidate.
- Disagreement with diversity: does avoiding redundant examples help?

Begin with random selection, S2 uncertainty, and disagreement. Expand only if the pilot shows an effect worth investigating.

A useful control is disagreement between two S2 models. If two optical models provide the same benefit, the mechanism may be ordinary model disagreement rather than complementary radar-optical information.

6. Separate the selector from the model benefiting from labels

If each selection strategy chooses a subset but also uses a different final architecture, improvements cannot be attributed to selection. Initially evaluate every selected subset using the same downstream model and training procedure.

Example: S1 and S2 teachers select chips; every subset is used to train the same S1-only CROMA configuration. This asks whether paired optical observations help allocate annotation effort for radar-only deployment.

Alternatively, use S2 as the downstream model and ask whether radar adds selection information beyond the stronger optical model. Pick one first. The radar deployment case connects to the observed S1 performance gap, but its practical value must be demonstrated.

7. Introduce weak labels as a second stage

Do not immediately combine active selection, weak supervision, fusion, and fine-tuning: their effects would be difficult to separate.

Once clean-label selection is understood:
- Give all strategies the same weak training data.
- Let each strategy choose scenes for manual correction.
- Replace weak labels with clean labels only for selected scenes.
- Measure whether the same correction budget produces different improvements.

Clean reference labels are not available for every weakly labelled chip. A retrospective simulation can use automatic labels on hand-labelled chips, if the required label sources exist or are generated consistently. A genuine experiment on the larger weak pool requires new annotation.

Distinguish annotation from correction: repairing an almost-correct mask may take much less time than drawing one from scratch. Without annotation-time measurements, report a chip budget rather than an unsupported claim about hours saved.

8. Evaluate the outcome that matters

The main figure is a learning curve:
- Horizontal axis: number of manually labelled chips.
- Vertical axis: performance on held-out flood events.
- Lines: annotation-selection strategies.
- Variation: repeated initial subsets and training seeds.

Useful outcomes:
- Water IoU at a fixed budget: segmentation quality with equal supervision.
- Flood-water recall and precision: missed inundation versus false alarms.
- Area under the learning curve: performance across several budgets.
- Labels needed to reach a predefined target: annotation efficiency.
- Per-event results: whether improvements are broad or driven by one event.

The existing PW/FW analysis can explain results, but its JRC-derived categorization is a proxy. Keep its limitations and aggregation conventions explicit.

9. What would count as a worthwhile finding?

Positive: cross-modal disagreement consistently selects more useful annotations than uncertainty and diversity, including on unseen events.

Conditional: it helps radar models or particular flood conditions, but not optical models generally.

Negative but informative: disagreement mainly captures acquisition mismatch or boundary noise, so annotating those scenes provides little learning benefit.

Diagnostic: optical uncertainty is equally effective, making the additional radar selection pipeline unnecessary.

Negative outcomes become scientifically useful when controls explain why. A result that merely fails to improve mIoU is weaker.

Agreement does not imply correctness: both sensors' models can inherit the same supervision bias. Independent review of agreement and disagreement cases would strengthen the study. Existing labels alone cannot prove that those labels are wrong.

10. Where novelty would actually need to come from

The disagreement formula, pseudo-labelling, and active-learning loop are established ideas. A possible contribution is convincing evidence about whether complementary sensors identify valuable annotations beyond single-sensor uncertainty and scene diversity, and under which conditions.

This remains a proposed question whose novelty needs verification, not a claim that no one has studied it. A focused review of cross-modal active learning for flood mapping must precede a novelty claim.

Practical next step: run a small feasibility study asking whether S1-S2 disagreement adds error-detection information beyond S2 uncertainty on development events. If it does not, investigate that finding before spending weeks on annotation selection. If it does, proceed to the controlled label-budget experiment.

The sequence is: diagnose the signal, test whether acting on it helps, then test whether the benefit transfers.